

Unit 1

Introduction to Business Intelligence (BI)

What is Business Intelligence (BI)?

Business Intelligence (BI) is the process of collecting, analyzing, and presenting business data to help companies make better decisions.

BI uses:

- **Data collection tools**
- **Reports and dashboards**
- **Charts and visualizations**
- **Data analysis techniques**

It converts raw data → useful information → smart business decisions.

Example:

A shop owner checks monthly sales data to know:

- **Which product sells the most**
- **Which month had low sales**
- **Which products need restocking**

Importance of BI

BI helps organizations:

- **Improve decision-making**
- **Increase sales and profit**
- **Understand customer behavior**
- **Identify market trends**

- Reduce business risks
- Improve operational efficiency

Real-life Examples

1. Retail Stores

- Track stock levels
- Know fast-selling products

2. Banks

- Detect fraud transactions
- Analyze customer spending habits

3. E-commerce Companies

- Recommend products to customers
- Analyze customer reviews and buying patterns

BI vs Analytics

BI (Business Intelligence)

Focuses on past and present data

Mainly reporting and dashboards

Answers “What happened?”

Used for monitoring business performance

Example

- BI: “Sales increased by 20% this month.”

Analytics

Focuses on future predictions and insights

Deep analysis and forecasting

Answers “Why did it happen?” and “What will happen?”

Used for improving strategies

- **Analytics: “Sales increased because of festival discounts and customer demand.”**
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Types of Analytics

1. Descriptive Analytics

Tells what happened in the past.

Example:

- **Monthly sales report**
- **Attendance report**
- **Website traffic summary**

Features:

- **Uses historical data**
 - **Simple charts and reports**
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2. Predictive Analytics

Predicts what may happen in the future using past data.

Example:

- **Predicting future sales**
- **Weather forecasting**
- **Predicting customer purchases**

Techniques Used:

- **Machine Learning**
- **Statistical models**

3. Prescriptive Analytics

Suggests what actions should be taken.

Example:

- **Best pricing strategy**
- **Product recommendations**
- **Route optimization for delivery**

Goal:

Help businesses choose the best decision.

How Businesses Use Data for Better Decisions

Businesses collect data from:

- **Sales systems**
- **Customer feedback**
- **Social media**
- **Websites**
- **Inventory systems**

They analyze this data to:

- **Understand customer needs**
- **Improve products/services**
- **Increase profits**
- **Reduce waste and losses**

Simple Example

If a supermarket notices that cold drinks sell more in summer, it can:

- **Increase stock during summer**
 - **Offer discounts**
 - **Improve sales planning**
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Data Architecture for BI

Data architecture refers to how data is collected, stored, processed, and used in BI systems.

1. Data Sources

Data sources are places from where data is collected.

Common Data Sources:

- **Databases**
- **Excel files**
- **Websites**
- **Mobile apps**
- **ERP systems**
- **CRM systems**
- **Social media platforms**

Example:

An online shopping company collects data from:

- **Customer orders**
- **Payment systems**

- **Delivery tracking**
-

OLTP vs OLAP

OLTP

Online Transaction Processing

Handles daily transactions

Fast insert/update/delete

Used in banking, shopping systems

Current data

Example

OLTP

- **ATM transactions**
- **Online shopping payment**

OLAP

- **Monthly sales analysis**
 - **Yearly profit comparison**
-

OLAP

Online Analytical Processing

Handles data analysis

Fast querying and reporting

Used in BI and reporting

Historical summarized data

Data Pipelines

A data pipeline is the process of moving data from source systems to BI tools.

Steps in Data Pipeline

- 1. Data Collection**
- 2. Data Cleaning**

3. Data Transformation

4. Data Storage

5. Data Analysis & Reporting

Example

Sales data from multiple stores is:

- **Collected**
 - **Cleaned**
 - **Combined into one database**
 - **Shown in dashboards**
-

Data Quality

Data quality means data should be:

- **Accurate**
- **Complete**
- **Consistent**
- **Reliable**

Poor Data Quality Problems

- **Wrong reports**
- **Bad business decisions**
- **Customer dissatisfaction**

Example

If customer phone numbers are incorrect, delivery services may fail.

Metadata

Metadata = “Data about data.”

It describes:

- **Data type**
- **Source**
- **Format**
- **Date created**
- **Meaning of data fields**

Example

In a student database:

- **Roll_No → Integer**
- **Student_Name → Text**
- **DOB → Date**

Metadata helps users understand and manage data properly.

Summary

Business Intelligence helps companies convert data into useful insights for better decision-making. BI uses tools like dashboards, reports, OLAP systems, and analytics techniques. Good data architecture, quality data, and proper data pipelines are essential for successful BI systems.

Unit 2

Making Decisions with Business Intelligence (BI)

Business Intelligence helps organizations make smart and data-driven decisions. Instead of guessing, companies use reports, charts, dashboards, and data analysis.

4 Steps of Decision Making in BI

1. Identifying the Problem

The first step is to understand what problem exists.

Examples:

- Sales are decreasing
- Stock is not selling
- Customer complaints are increasing

BI Role:

BI tools show reports and dashboards that help detect problems quickly.

Example:

A store owner notices from a sales chart that laptop sales dropped in March.

2. Finding Possible Options

After identifying the problem, businesses think about different solutions.

Example Options:

- Give discounts
- Advertise more
- Increase online promotion
- Change product pricing

BI Role:

BI compares past data and trends to suggest better choices.

3. Choosing the Best Option

Businesses analyze data to select the most effective solution.

Example:

If reports show discounts increased sales last year, the company may choose discounts again.

BI Tools Used:

- Charts
 - Dashboards
 - Tables
 - KPI reports
-

4. Taking Action

The final step is implementing the selected decision.

Example:

- Launch discount offers
- Increase stock
- Stop low-selling products

After action, BI tools again monitor results to check improvement.

Using Charts and Tables in Decision Making

Charts and tables make data easy to understand.

Types of Charts Used in BI

Chart Type Purpose

Bar Chart Compare products or sales

Line Chart Show trends over time

Pie Chart Show percentage distribution

Table Detailed numeric data

Example: Product Sales Decision

Product Monthly Sales

Shoes 500

Bags 200

Watches 100

Decision:

- Sell more shoes
- Reduce watch stock
- Promote bags with offers

BI helps businesses make these decisions quickly using visual reports.

Data Warehousing Concepts

What is a Data Warehouse (DW)?

A **Data Warehouse** is a centralized storage system that stores large amounts of historical business data for analysis and reporting.

It collects data from different sources and organizes it for BI and analytics.

Features:

- Stores historical data
- Supports decision making
- Read-only and analysis-oriented
- Large and integrated database

Example:

A supermarket stores years of sales data in a warehouse to analyze customer buying patterns.

Data Mart

A **Data Mart** is a smaller part of a data warehouse focused on a specific department or business area.

Examples:

- Sales Data Mart
- HR Data Mart
- Finance Data Mart

Difference Between DW and Data Mart**Data Warehouse**

Large centralized system

Used by whole organization

Stores all business data

Data Mart

Smaller departmental system

Used by one department

Stores specific data

Data Warehouse Architecture

A basic DW architecture has three layers:

1. Data Sources Layer

Data comes from:

- Databases
 - ERP systems
 - CRM systems
 - Excel files
-

2. ETL Layer

Data is:

- Extracted
- Cleaned
- Transformed
- Loaded into warehouse

3. Presentation Layer

Users access:

- Reports
 - Dashboards
 - BI tools
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Fact Table and Dimension Table

Fact Table

Contains:

- Numerical business data
- Measurements or metrics

Examples:

- Sales amount
- Quantity sold
- Profit

Example Table

Product_ID	Customer_ID	Sales
------------	-------------	-------

101	1	5000
-----	---	------

Dimension Table

Contains descriptive information about facts.

Examples:

- Product details
- Customer details
- Time details

Example Table

Product_ID	Product_Name	Category
------------	--------------	----------

101	Shoes	Footwear
-----	-------	----------

Star Schema

In a **Star Schema**, one central fact table is connected directly to multiple dimension tables.

Structure:

- Fact table at center
- Dimension tables around it

It looks like a star.

Advantages:

- Simple design
 - Fast query performance
 - Easy to understand
-

Snowflake Schema

A **Snowflake Schema** is an extension of star schema where dimension tables are further divided into sub-dimension tables.

Features:

- More normalized
 - Reduces data redundancy
 - More complex structure
-

Difference Between Star and Snowflake Schema

Star Schema	Snowflake Schema
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Simple structure	Complex structure
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Less normalization	More normalization
--------------------	--------------------

Faster queries	Slightly slower
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More redundancy	Less redundancy
-----------------	-----------------

ETL & Data Integration

What is ETL?

ETL stands for:

1. **Extract**
2. **Transform**
3. **Load**

It is the process of moving data from source systems into a data warehouse.

1. Extract

Data is collected from multiple sources.

Sources:

- Databases
- CSV files
- Websites
- ERP systems

Example:

Collecting sales data from different branch stores.

2. Transform

Data is cleaned and converted into proper format.

Tasks:

- Remove duplicate data
- Correct errors
- Standardize formats
- Calculate values

Example:

Changing dates into same format.

3. Load

Cleaned data is loaded into the data warehouse.

Types:

- Full Load
 - Incremental Load
-

Data Integration

Data integration means combining data from multiple systems into one unified view.

Purpose:

- Better analysis
 - Consistent reporting
 - Single source of truth
-

Data Integration Approaches

1. Manual Integration

Data is combined manually using spreadsheets or files.

Limitation:

- Time-consuming
 - Error-prone
-

2. Middleware Integration

Special software connects different systems automatically.

Example:

Enterprise Application Integration (EAI)

3. ETL-Based Integration

Most common approach in BI.

Data is extracted, transformed, and loaded into warehouse.

4. Real-Time Integration

Data updates instantly.

Example:

Live stock market dashboards

Summary

Business Intelligence supports decision-making through reports, charts, and analytics. Data warehouses store large historical data for analysis, while ETL processes help collect and integrate data from different sources. Concepts like fact tables, dimension tables, star schema, and snowflake schema form the foundation of modern BI systems.

Unit 3

Basic Analytics

Analytics helps businesses understand data and make better decisions.

It answers questions like:

- What happened?

- What may happen?
 - What should we do?
-

Types of Analytics

1. Descriptive Analytics

Descriptive analytics explains **what happened in the past**.

Example:

- Last month's sales report
- Number of products sold
- Website visitors in a week

Purpose:

To summarize historical data using:

- Reports
- Charts
- Dashboards
- Tables

Example:

A shop finds that it sold **500 shoes in April**.

2. Predictive Analytics

Predictive analytics estimates **what might happen in the future** using past data.

Example:

- Forecasting next month's demand
- Predicting customer purchases
- Predicting future profits

Techniques Used:

- Historical trends
- Statistical models
- Machine Learning

Example:

If shoe sales increase every summer, the company predicts higher sales next month too.

3. Basic Prescriptive Analytics

Prescriptive analytics suggests **what action should be taken**.

Example Questions:

- Should we increase stock?
- Should we reduce prices?
- Which product should be promoted?

Example:

If demand is expected to rise, the business may increase inventory.

Finding Patterns in Small Datasets

Businesses analyze small datasets to identify useful patterns.

Methods Used

1. Average

Average helps find the normal or expected value.

Formula

$$\text{Average} = \frac{\text{Sum of Values}}{\text{Number of Values}}$$

Example:

Sales for 5 days:

- 100

- 120
- 140
- 110
- 130

Average Sales:

$$= (100 + 120 + 140 + 110 + 130) \div 5$$

$$= 120$$

Use:

Businesses use averages to:

- Estimate demand
- Compare performance
- Plan stock

Using Charts

Charts make patterns easy to understand.

Chart Type Purpose

Bar Chart Compare products

Line Chart Show trends over time

Pie Chart Show percentages

Example:

A bar chart may show:

- Shoes = Highest sales
- Watches = Lowest sales

This helps businesses decide what to sell more or less.

Grouping Data

Grouping means organizing similar data together.

Example:

Products grouped by category:

- Electronics
- Clothing
- Food

Benefits:

- Easier analysis
 - Better reporting
 - Faster decision-making
-

Facts, Dimensions, and Measures

These are important concepts in BI and data warehousing.

1. Facts

Facts are numerical values or measurable business data.

Examples:

- Sales amount
- Profit
- Quantity sold

Fact Table Example

Product_ID	Store_ID	Sales
------------	----------	-------

101	1	5000
-----	---	------

Facts are usually stored in **fact tables**.

2. Dimensions

Dimensions describe the facts.

Examples:

- Product name
- Customer name
- Date
- Location

Dimension Table Example

Product_ID	Product_Name	Category
------------	--------------	----------

101	Shoes	Footwear
-----	-------	----------

Dimensions help answer:

- Which product?
- Which customer?
- Which location?

3. Measures

Measures are values used for calculations and analysis.

Examples:

- Total Sales
- Average Profit
- Number of Orders

Measures are usually numeric and stored in fact tables.

Hierarchy in BI

A hierarchy shows levels of data organization.

Example: Time Hierarchy

- Year

- Quarter
 - Month
 - Day

Example: Location Hierarchy

- Country
 - State
 - City

Hierarchies help users drill down into detailed data.

Keys in Data Warehousing

Keys help connect tables.

1. Primary Key

A unique identifier in a table.

Example:

Product_ID

Product_ID	Product_Name
------------	--------------

101	Shoes
-----	-------

2. Foreign Key

A key used to connect one table to another.

Example:

Product_ID in fact table links to dimension table.

Designing with Example

Example: Sales Data Warehouse Design

Fact Table: Sales_Fact

Product_ID	Time_ID	Store_ID	Sales
------------	---------	----------	-------

101	1	10	5000
-----	---	----	------

Product Dimension

Product_ID	Product_Name	Category
------------	--------------	----------

101	Shoes	Footwear
-----	-------	----------

Time Dimension

Time_ID	Month	Year
---------	-------	------

1	April	2026
---	-------	------

Store Dimension

Store_ID	City	State
----------	------	-------

10	Delhi	Delhi
----	-------	-------

How It Helps

Businesses can now analyze:

- Sales by product
- Sales by city
- Monthly profit
- Best-selling categories

This structure improves reporting and decision-making.

Summary

Basic analytics helps organizations understand past performance, predict future outcomes, and suggest actions. Concepts like facts, dimensions, measures, hierarchies, and keys are essential for designing BI systems and analyzing business data effectively.

Unit 4

Handling Data in Business Intelligence (BI)

Handling data means collecting, cleaning, organizing, and storing data properly so it can be easily analyzed and used for decision-making.

Using Excel or Simple Tools to Gather Information

Organizations collect data using simple tools such as:

- Microsoft Excel
- Google Sheets
- CSV files
- Forms and surveys
- Databases

Example:

A shop owner records:

- Product sales
- Customer details
- Stock quantity

using Excel sheets.

Common Tasks in Excel for BI

1. Data Entry

Entering information in rows and columns.

Product Sales

Shoes 5000

Bags 3000

2. Sorting Data

Arranging data:

- Alphabetically
- Highest to lowest
- Date-wise

Example:

Sort products by highest sales.

3. Filtering Data

Showing only required information.

Example:

Display only products with sales above ₹5000.

4. Using Formulas

Excel formulas help perform calculations.

Common Functions:

- SUM()
- AVERAGE()
- COUNT()
- MAX()
- MIN()

Example:

Calculate total monthly sales.

Removing Mistakes or Duplicates

Before analysis, data must be cleaned.

This process is called **Data Cleaning**.

Common Data Problems

Problem	Example
Duplicate records	Same customer entered twice
Missing values	Blank phone number
Wrong spelling	“Delhii” instead of “Delhi”
Incorrect formats	Different date formats

Simple Ways to Clean Data**1. Remove Duplicates**

Excel provides a “Remove Duplicates” option.

Benefit:

Avoids counting same data multiple times.

2. Correct Errors

Fix spelling mistakes and incorrect entries.

Example:

Change:

- “Mumbay” → “Mumbai”

3. Fill Missing Values

Replace blanks with correct information if available.

4. Standardize Formats

Keep data in same format.

Example:

Use one date format:

- DD/MM/YYYY
-

Storing Data Properly

Data should be organized so it is:

- Easy to search
 - Easy to update
 - Easy to analyze
-

Good Data Storage Practices

1. Use Proper Table Structure

Customer_ID Name City

101 Ali Delhi

2. Use Unique IDs

Each record should have a unique identifier.

Example:

- Product_ID
- Customer_ID

3. Keep Backup

Always maintain backup copies of important data.

4. Organize Files Properly

Use clear file names and folders.

Example:

- Sales_2026.xlsx
 - Customer_Data.xlsx
-

BI Tools Landscape

BI tools help organizations:

- Analyze data
- Create reports
- Build dashboards
- Visualize trends

Popular BI tools include:

- Microsoft Power BI
 - Tableau
 - Qlik Sense
-

1. Microsoft Power BI

A popular BI tool developed by Microsoft.

Features:

- Interactive dashboards
- Data visualization

- Report generation
- Easy Excel integration

Used For:

- Sales analysis
 - KPI tracking
 - Business reporting
-

2. Tableau

A powerful visualization tool used for creating charts and dashboards.

Features:

- Drag-and-drop interface
- Interactive graphs
- Fast analytics

Best For:

- Data visualization
 - Business reporting
 - Trend analysis
-

3. Qlik Sense

A BI and analytics platform known for associative data analysis.

Features:

- Smart search
 - Interactive dashboards
 - Self-service analytics
-

Dashboards vs Reports

Dashboard

Report

Visual summary of data Detailed information

Interactive

Mostly static

Real-time monitoring

Historical analysis

Quick insights

Detailed explanation

Dashboard Example

A sales dashboard may show:

- Total sales
- Top-selling products
- Monthly growth
- Customer count

using charts and KPIs.

Report Example

A monthly report may contain:

- Complete sales records
- Product-wise details
- Employee performance

KPIs (Key Performance Indicators)

KPIs are measurable values used to check business performance.

Examples:

- Total Sales
- Profit Margin

- Customer Satisfaction
- Website Traffic

Example:

If a company target is ₹10 lakh sales per month, actual sales become a KPI.

Metrics

Metrics are measurements used for analysis.

Examples:

- Number of customers
- Revenue
- Orders completed
- Delivery time

Difference Between KPI and Metric

KPI	Metric
Measures business goal achievement	Measures general performance
Strategic	Operational

Scorecards

Scorecards compare:

- Targets
- Actual performance

They help management track progress.

Example Scorecard

KPI	Target	Actual
Monthly Sales	₹5,00,000	₹4,80,000
Customer Growth	10%	8%

Best Practices in BI

1. Keep Data Clean

Incorrect data gives wrong insights.

2. Use Simple Visualizations

Avoid overly complex charts.

3. Focus on Important KPIs

Track only meaningful business metrics.

4. Update Data Regularly

Outdated data can lead to poor decisions.

5. Ensure Data Security

Protect sensitive business information.

6. Make Dashboards Easy to Understand

Use:

- Clear labels
- Proper colors
- Simple layout

Summary

Handling data properly is an important part of BI. Tools like Excel help gather and clean data, while BI tools such as Power BI, Tableau, and Qlik help analyze and visualize information.

Dashboards, reports, KPIs, metrics, and scorecards support businesses in tracking performance and making smarter decisions.